

BUILDTRACK

Construction Project Management & Site Monitoring Platform

MILESTONE 1 DOCUMENTATION

Modules 1–3: Authentication, Administration & Project Manager

Document Item	Details
Project	BuildTrack – Construction Project Management & Site Monitoring Platform
Milestone	Milestone 1
Modules Covered	Module 1 – Authentication; Module 2 – Administration; Module 3 – Project Manager
Technology	Angular frontend; Python FastAPI backend; SQLAlchemy/database layer; JWT-based authentication
Document Type	Technical Project / Internship Documentation
Evidence Basis	Current BuildTrack project source code and reference milestone specification

Version: 1.0

1. Milestone Overview

Milestone 1 establishes the core access, administration and project-management foundation of BuildTrack. The implemented frontend contains dedicated authentication pages, an administrator workspace with user and project management screens, and a Project Manager workspace containing project monitoring and planning views. The backend contains FastAPI endpoints and database models supporting user authentication and project-related operations.

Module	Purpose	Main Implemented Areas
Module 1 – Authentication	Control user entry into BuildTrack and establish authenticated sessions.	Login, Remember Me, JWT token storage, role-based redirection, registration route, password-reset route.
Module 2 – Administration	Provide administrators with centralized control over users, projects and administrative monitoring.	Admin dashboard, user directory/detail/add/edit/view screens, project listing/add/edit/view/detail screens, analytics, reports and audit-log routes.
Module 3 – Project Manager	Give project managers a consolidated view of projects, milestones, site progress, resources, workforce, procurement and budget information.	Dashboard, My Projects, Project Details, Project Schedule, Milestones, Site Engineers, Contractors, Site Progress, Resource Allocation, Workforce, Procurement Requests, Budget Tracking, Reports, Notifications and Profile.

2. Milestone Objectives

- Establish the authentication entry point and authenticated role-based navigation.
- Provide administrator-facing screens for user and project administration.
- Provide a Project Manager workspace for project-level monitoring and planning.
- Connect the frontend structure with backend authentication and project APIs where implemented.
- Create a clear navigation foundation for the later construction-management modules.

3. System Architecture Used in Milestone 1

The project uses Angular for the frontend and Python FastAPI for the backend. The frontend uses Angular standalone components and route-based navigation. The backend uses FastAPI, SQLAlchemy and a database session layer. JWT tokens are generated during login and are used by the application to identify the authenticated user. The frontend stores the access token and selected user information in localStorage and redirects the user according to the returned role.

Layer	Observed Implementation
Frontend	Angular 22 project structure with standalone components, Angular Material components and route configuration.
Routing	Angular Routes with dedicated authentication, administrator, Project Manager and other module paths.
Backend	Python FastAPI application with routers for project, milestone, worker, contractor, site-engineer, schedule and project-closure operations.
Authentication	JWT access token creation and verification; password hashing/verification using Passlib bcrypt.
Database Access	SQLAlchemy engine/session configuration and ORM models.
Authorization	Angular authGuard/roleGuard are defined and used on protected module routes such as Site Engineer; login response also supplies the user's role for redirection.

4. Module 1 – Authentication

4.1 Objective

The Authentication module provides the entry point into BuildTrack and supports credential-based login, session persistence preferences and role-specific navigation. The current frontend exposes Login, Register and Reset Password routes.

4.2 Features

Feature	How it works in BuildTrack	Role / Output
Login	The user enters email and password and selects Login. The Angular component sends email, password and remember_me to AuthService. A successful response stores the access token, role, full name and expiry time.	All authenticated users / authenticated session and role-specific navigation.
Remember Me	The login request includes the remember_me value. The backend creates a longer-lived token when this option is enabled.	User / extended token lifetime when enabled.
Password Visibility	The login page provides a visibility control that toggles the password field between hidden and visible states.	User / easier credential entry.
Role-based Redirect	After successful authentication, the frontend checks the returned role and redirects Administrator to the Admin Dashboard, Project Manager to the Project Manager Dashboard, and Site Engineer to the Site Engineer Dashboard.	System / role-specific workspace.
Registration	A /register route and Register component are present. The current frontend component is a placeholder screen, while the backend exposes a /register endpoint that validates email and employee-ID uniqueness, confirms passwords and creates a user.	New user / registration workflow is partially implemented.
Password Reset	A /reset-password route and component are present. The current frontend component is a placeholder. The backend exposes a /forgot-password endpoint for initiating password recovery.	User / recovery workflow is partially implemented.

4.3 Authentication Workflow

Step	User Action	System Response	Result
1	Open BuildTrack.	Default route redirects to /login.	Login screen is displayed.
2	Enter email and password.	Angular binds the values to the login component.	Credentials are prepared.
3	Optionally select Remember Me.	remember_me is included in the request.	Token lifetime preference is passed to backend.
4	Click Login.	AuthService sends the login request to the backend.	Authentication is attempted.
5	Credentials are valid.	Backend verifies the user and returns JWT, role, name and expiry information.	Authenticated response is received.
6	Frontend stores authentication data.	access_token, role, full_name and expires_at are stored in localStorage.	Client-side session information is available.
7	System checks role.	Router navigation is selected according to role.	User enters the appropriate workspace.

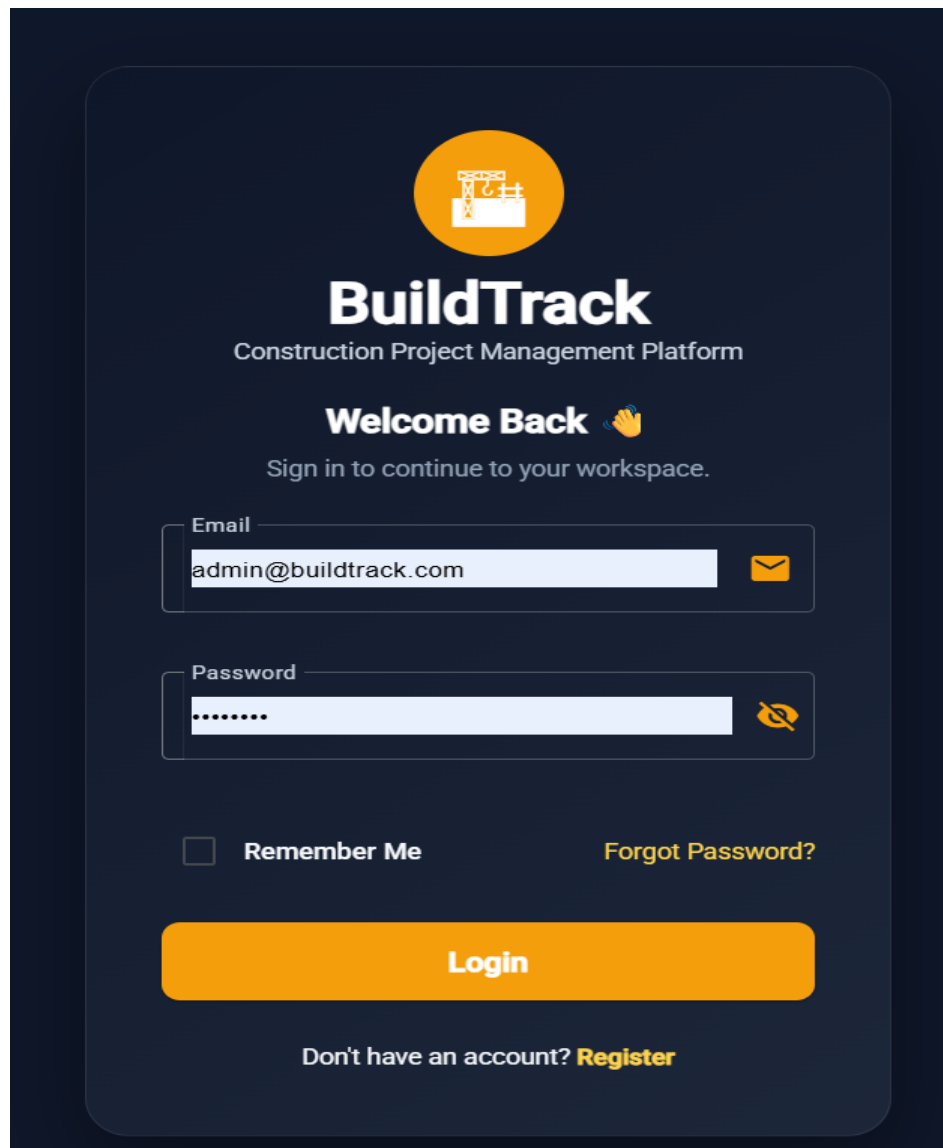


Figure 1: BuildTrack Login Page

Authentication Login Screen

Screenshot placement: capture the actual /login page showing the BuildTrack branding, email and password fields, Remember Me option, Login button and Register link.

4.4 Technical Implementation

- Angular Login component is implemented as a standalone component.
- Angular FormsModule and Material form/input/button/icon/checkbox modules are used in the login page.
- AuthService is called through login(loginData).
- Successful authentication stores access_token, role, full_name and expires_at in localStorage.
- FastAPI provides /login and /register endpoints and uses JWT creation/verification.
- Passwords are hashed and verified using a Passlib bcrypt context.
- The backend checks whether the account is verified before allowing login.

4.5 Module Outcome

The authentication foundation is present and the login workflow is connected to the backend authentication service. The Register and Reset Password frontend screens are currently placeholders and should be completed before describing those two UI workflows as fully operational.

5. Module 2 – Administration

5.1 Objective

The Administration module provides an administrator workspace for monitoring the overall system and managing users and projects. The route configuration includes an AdminLayout with dedicated dashboard, user-management and project-management screens, together with analytics, audit-log and report routes.

5.2 Administrator Dashboard

The Admin Dashboard presents high-level operational indicators and supporting project information. The implemented page contains KPI cards for Active Projects, Workforce, Budget and Procurement, followed by project progress, a focus card and a live project table.

Dashboard Area	Displayed Information
Active Projects	24, with a weekly change indicator.
Workforce	356, with an attendance indicator.
Budget	₹2.4 Cr, with utilization indicator.
Procurement	8 pending approvals.
Project Progress	Visual project-progress component.
Focus / Live Projects	Supporting focus information and a live project table.

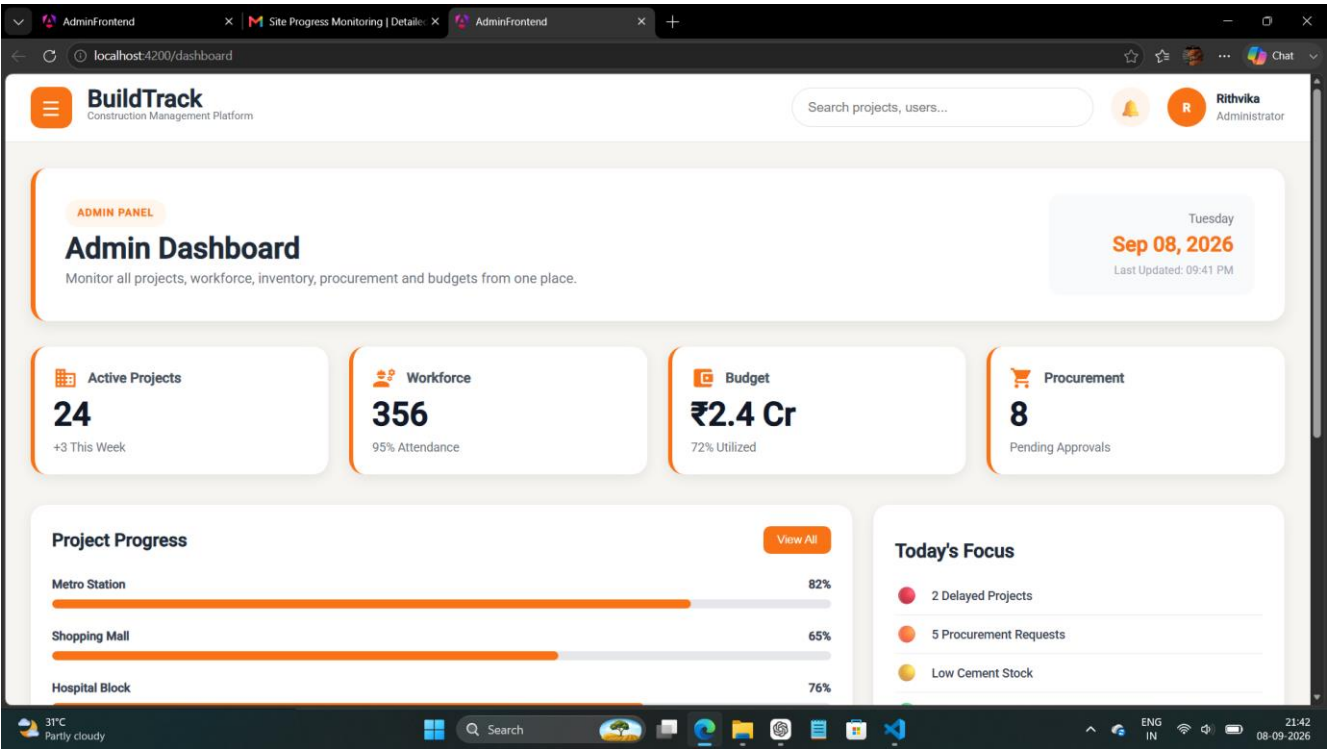


Figure 2: BuildTrack Administrator Dashboard

Administrator Dashboard

5.3 User Management

The user-management area contains a User Directory with search capability and administrative actions. Separate screens are available for adding, viewing and editing users. The Add User form contains personal information, employment details and account details.

Screen	Observed Functionality
User Directory / User Details	Search users by name, employee ID or role; display user records with ID, Employee, Role, Department, Status and Actions.
Add User	First Name, Last Name, Email Address, Phone Number, Gender, Role, Department, Employee ID, Status, Password and Confirm Password fields.
View User	Displays personal and employment information including name, email, phone, gender, employee ID, department, role, status and joining date.
Edit User	Provides editable user information and account/status fields with navigation back to user details.

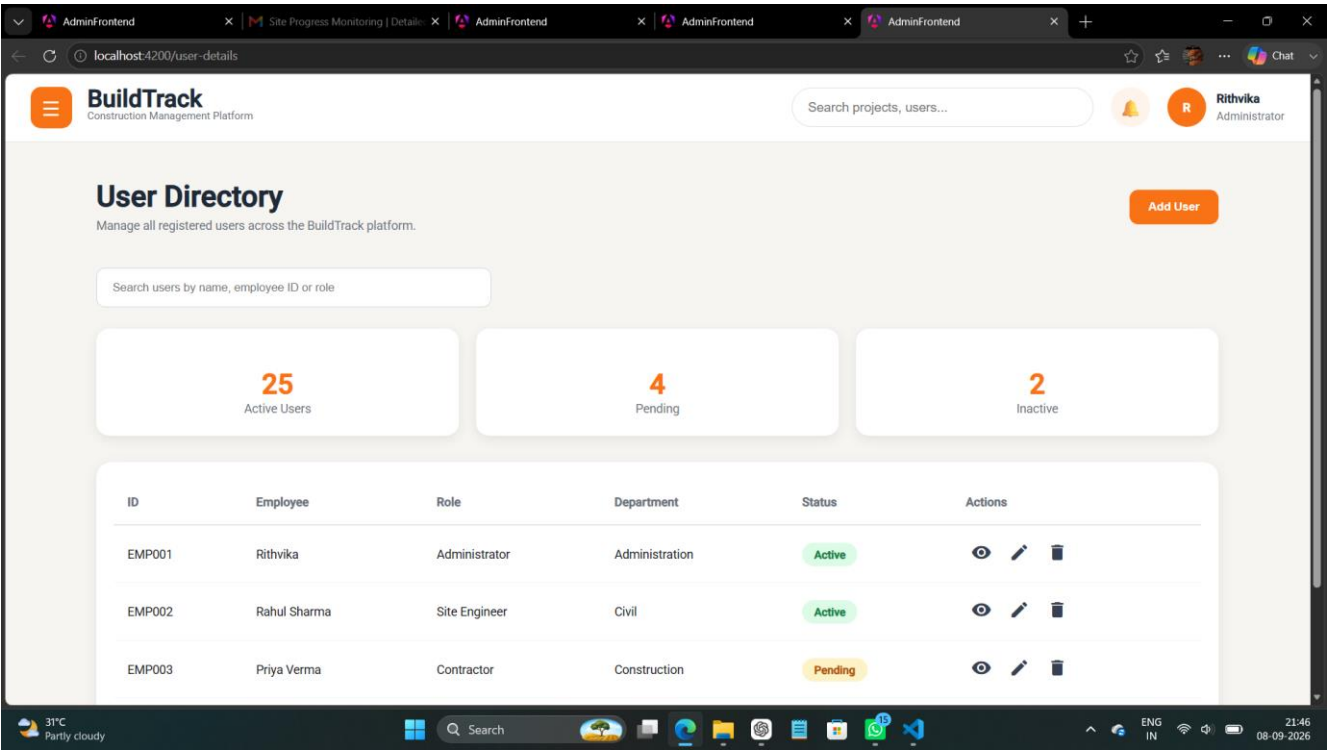


Figure 3: BuildTrack Administrator – User Directory

User Directory

Figure 4: BuildTrack Administrator – Add New User

Add User Form

Figure 5: BuildTrack Administrator – View User

View User

Figure 6: BuildTrack Administrator – edit User

edit User

BuildTrack
Construction Management Platform

Search projects, users...

Edit User
Update the user's information and account details.

Personal Information

First Name: Rithvika

Last Name: V

Email Address: rithvika@gmail.com

Phone Number: +91 9876543210

Gender: Female

Employment Details

Role:

Department:

5.4 Project Management

The administrator project-management area provides project listing and project information screens. The Add and Edit Project forms cover project identity, client and location information, management assignments, status, budget, dates and description. The View Project screen displays financial, timeline, progress and team-related information.

Screen	Observed Functionality
Project Management / Projects	Project listing/summary area with project-management navigation.
Add New Project	Project Name, Project ID, Client Name, Location, Project Manager, Site Engineer, Status, Budget, Start Date, Expected End Date and Description.
Project Details	Dedicated project-detail view.

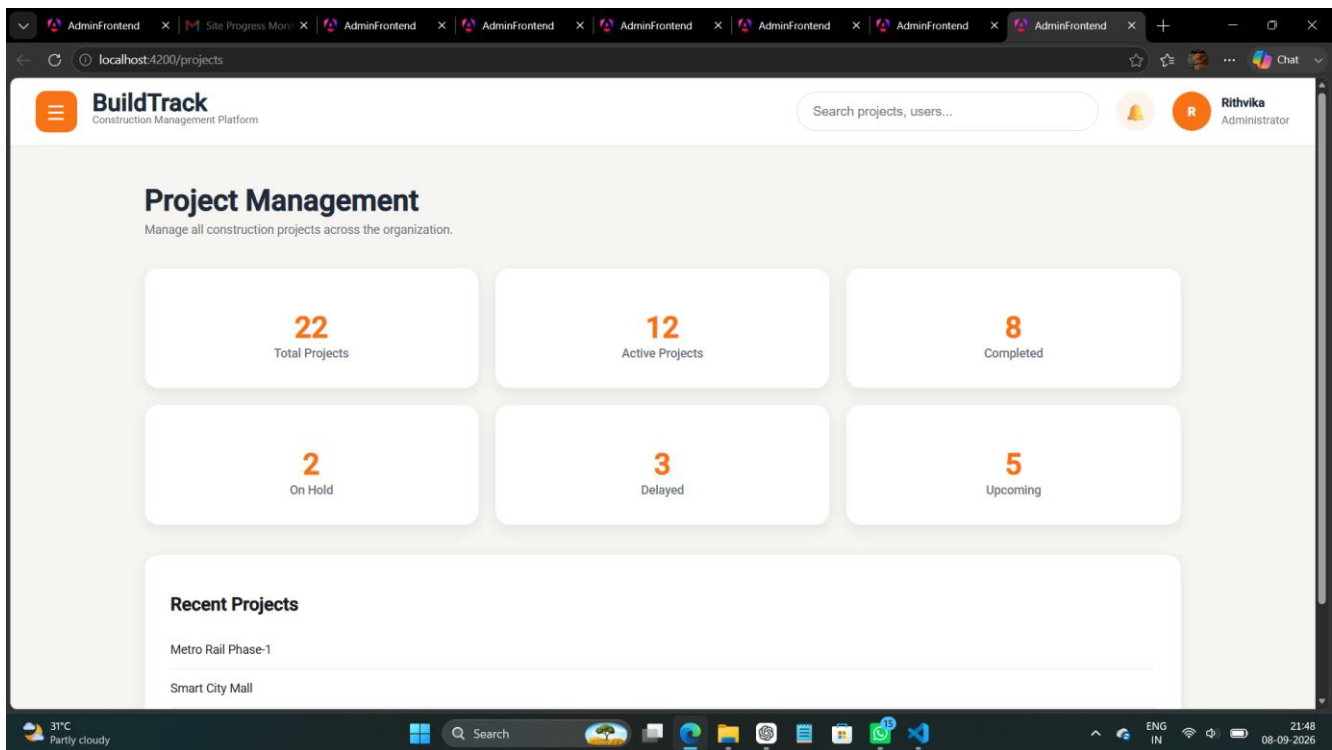


Figure 6: BuildTrack Administrator – Project Management

Project Management

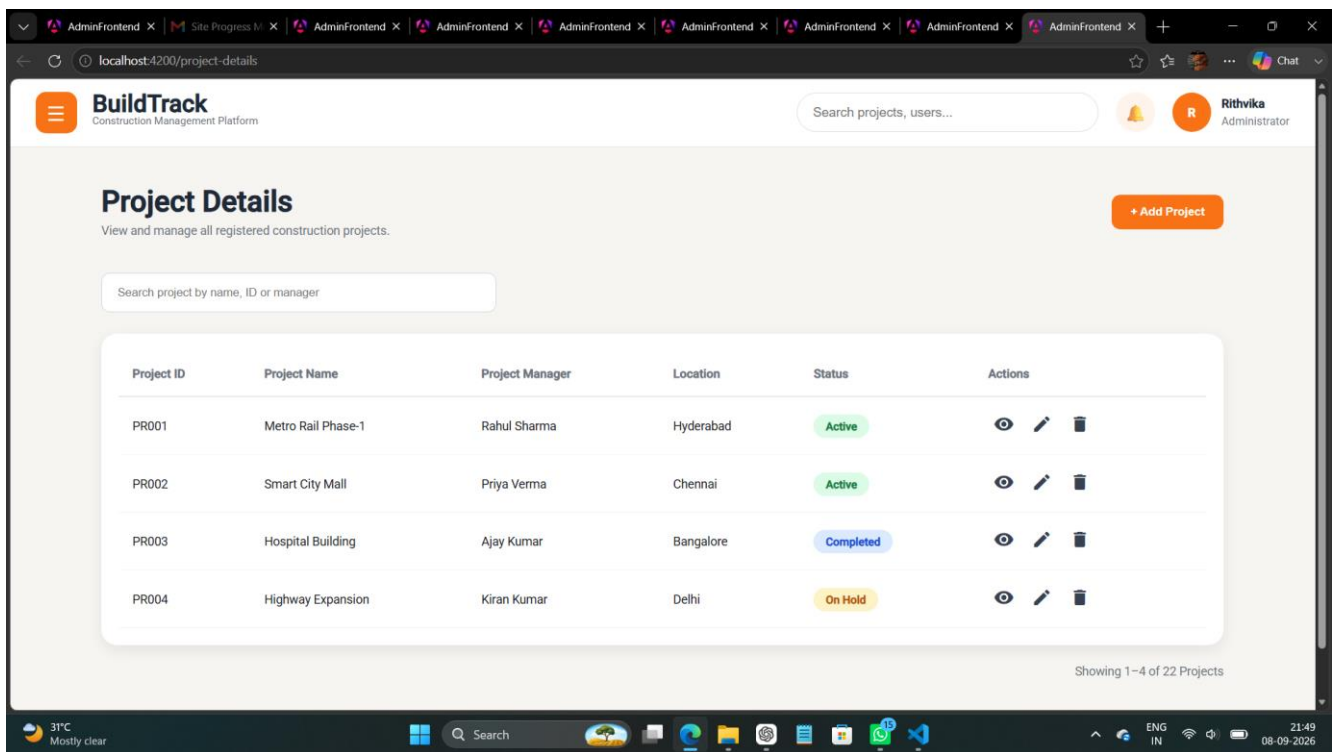


Figure 7: BuildTrack Administrator – Project details

Project details

Figure 8: BuildTrack Administrator – Add New Project

Add New Project

5.5 Administrative Navigation

The Admin route configuration includes dashboard, add/view/edit user, project listing/details/add/edit/view, analytics, audit logs and reports. These routes establish the navigation foundation for administrative monitoring and later integration of broader system functions.

5.6 Administrator Workflow

User Action	System Response	Result
Open Admin Dashboard	AdminLayout loads the dashboard component.	Administrative overview is displayed.
Open User Directory	User-management screen displays user records and actions.	Administrator can inspect the user directory.
Select Add User	Add User form opens.	Administrator can enter user and role information.
Select View/Edit User	Corresponding user screen opens.	Administrator can inspect or modify user information.
Open Project Management	Project listing/detail navigation is displayed.	Administrator can inspect projects.
Select Add Project	Project form or detail screen opens.	Project information can be entered, reviewed or edited.

5.7 Module Outcome

The administrative foundation is implemented at the frontend level with dashboard, user-management and project-management screens. The screens provide the required navigation and data-entry structures. The exact level of backend persistence should be demonstrated from the running application before claiming every UI action as fully integrated.

6. Module 3 – Project Manager

6.1 Objective

The Project Manager module provides a centralized workspace for monitoring assigned projects and related construction activities. The implemented dashboard brings project progress, milestones, site progress, resources, workforce, procurement, budget, delays, analytics, reports, notifications and profile information into one workspace.

6.2 Project Manager Dashboard

The Project Manager Dashboard displays four top-level dashboard cards: Assigned Projects, Completed Milestones, Delayed Activities and Budget Utilization. The page then presents sections for Assigned Projects, Project Milestones, Site Progress, Resource Allocation, Workforce Summary, Procurement Requests, Budget Tracking, Delayed Activities, Project Analytics, Reports, Recent Updates, Notifications and Profile.

Dashboard Section	Observed Information
Summary Cards	Assigned Projects, Completed Milestones, Delayed Activities and Budget Utilization.
Assigned Projects	Project, client, status and completion percentage.
Project Milestones	Milestone/task, completion percentage, deadline and status.
Site Progress	Site, engineer, progress and status.
Resource Allocation	Resource, assigned project and availability/allocation status.
Workforce Summary	Team, worker count and supervisor.
Procurement Requests	Material, supplier and request status.
Budget Tracking	Project, budget, spent and remaining amount.
Delayed Activities	Activity, reason and delay duration.
Project Analytics	Project completion, resource utilization, budget utilization and on-time delivery.
Reports / Notifications / Profile	Navigation/content areas for reporting, updates and Project Manager profile.

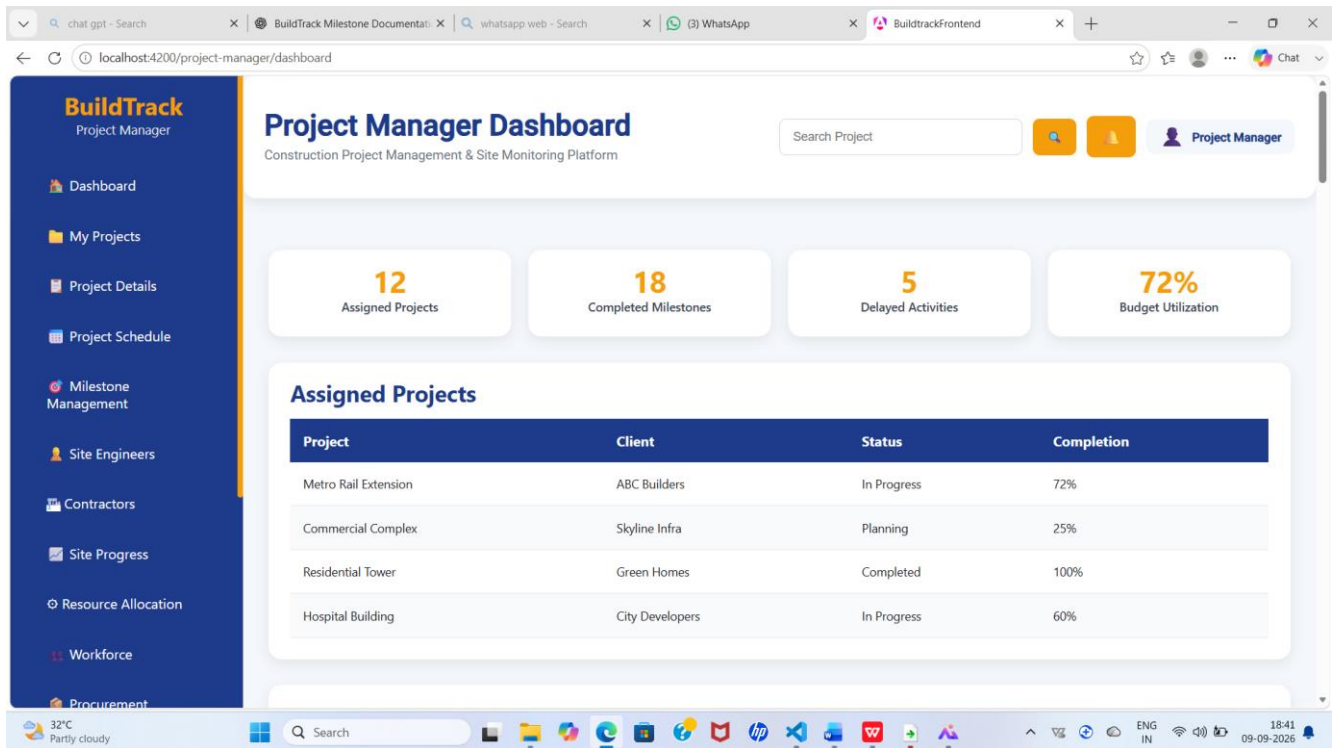


Figure 9: BuildTrack Project Manager Dashboard

Project Manager Dashboard

6.3 My Projects

The My Projects screen provides project-level filtering and summary indicators. It includes a search field for project or client and a project table containing Project Name, Code, Client, Location, Status, Priority, Budget, Completion, Start Date, End Date and Actions.

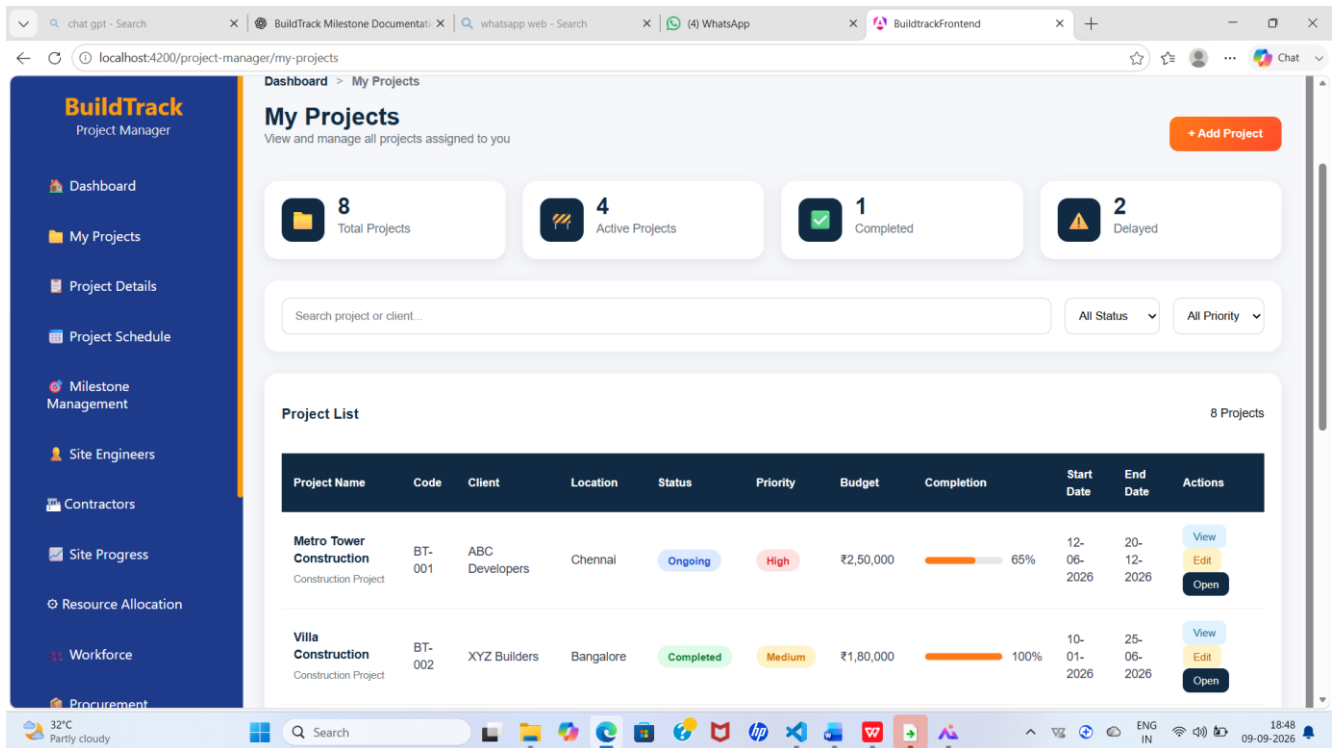


Figure 10: BuildTrack Project Manager – My Projects

My Projects

6.4 Project Details

The Project Details screen provides basic project information including project code, client, location, budget, start/end dates, status and priority. It also includes project progress, milestone details, site engineers, contractors, resources and materials.

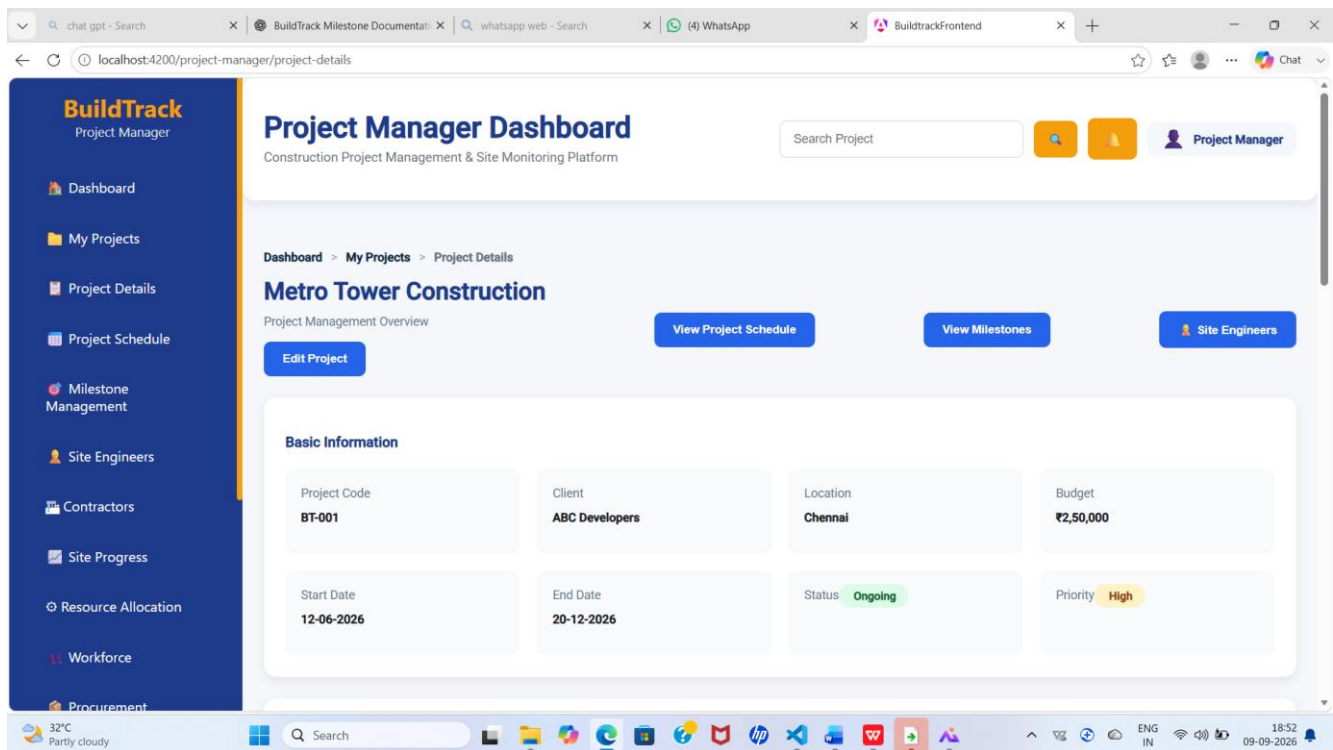


Figure 11: BuildTrack Project Manager – Project Details

Project Details

6.5 Project Schedule

The Project Schedule page provides a project timeline and an activity schedule. The timeline represents stages such as Foundation, Structure, Electrical, Plumbing, Painting, Finishing, Inspection and Handover. The activity table contains task name, start date, end date, duration, assigned engineer, status and actions.

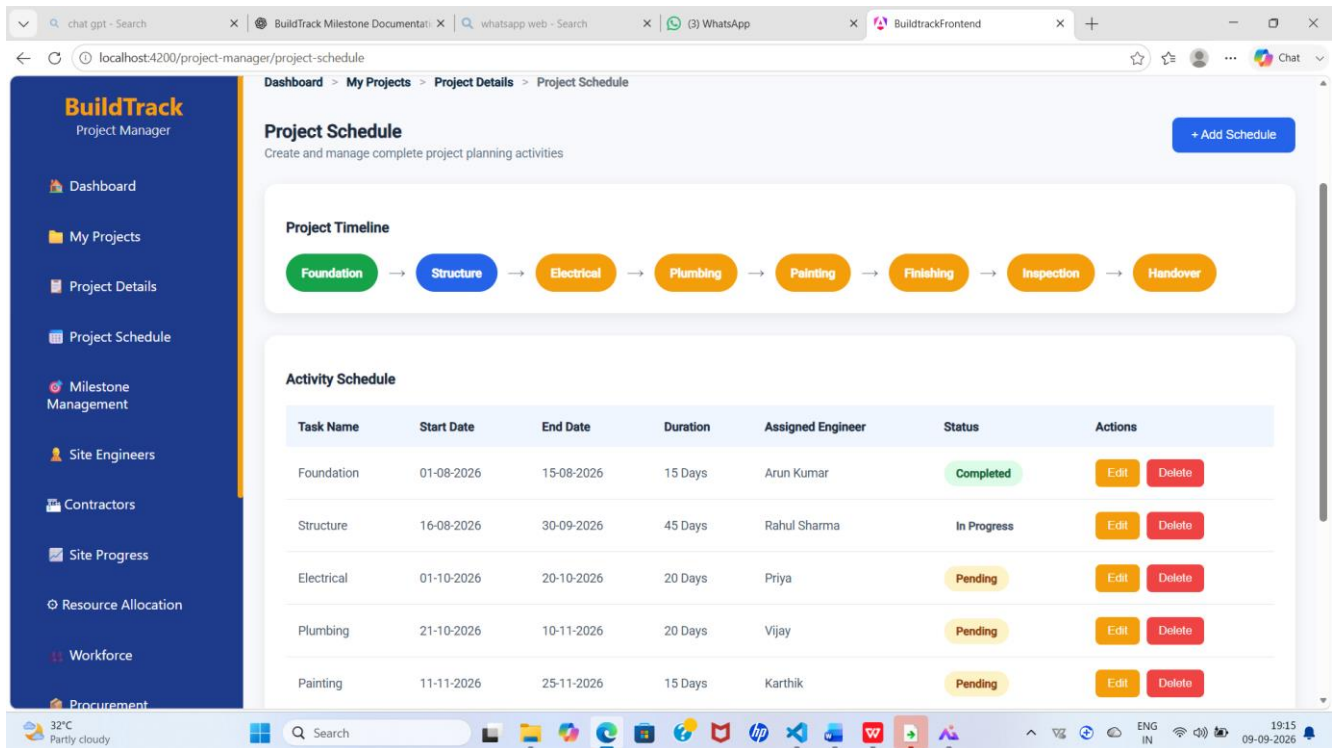


Figure 12: BuildTrack Project Manager – Project Schedule

Project Schedule

6.6 Milestone Management

The Milestones page provides a milestone timeline and a milestone-details table. The displayed milestone workflow includes stages such as Foundation Complete, Structure Complete, Electrical Complete, Plumbing Complete, Interior Complete, Final Inspection and Handover. The table records planned date, actual date, status, delay and remarks, with an action to complete a milestone.

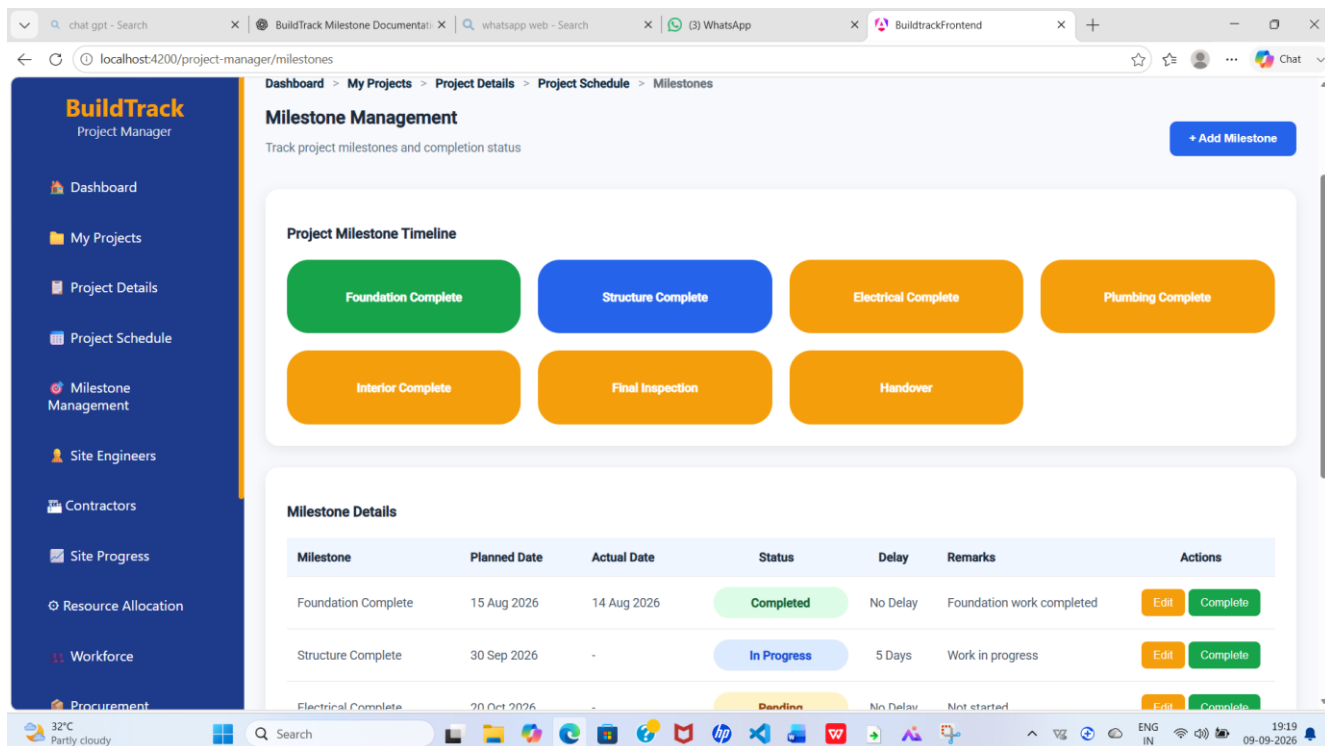


Figure 13: BuildTrack Project Manager – Milestone Management

Milestone Management

6.7 Supporting Project Manager Pages

Page	Purpose
Site Engineers	View project-associated site engineers and assignment information.
Contractors	View contractor/company and specialization information.
Site Progress	Monitor site-level progress and status.
Resource Allocation	View allocated resources against projects.
Workforce	View workforce/team information.
Procurement Requests	View material/supplier procurement request information.
Budget Tracking	View project budget, spending and remaining amount.
Reports	Access Project Manager reporting area.
Notifications	View Project Manager notifications/updates.
Profile	Display Project Manager profile information.

6.8 Site Engineers

The Site Engineers page allows the Project Manager to view site engineers associated with the project. It provides information about assigned engineers and their project responsibilities. This helps the Project Manager monitor and coordinate site-level activities effectively.

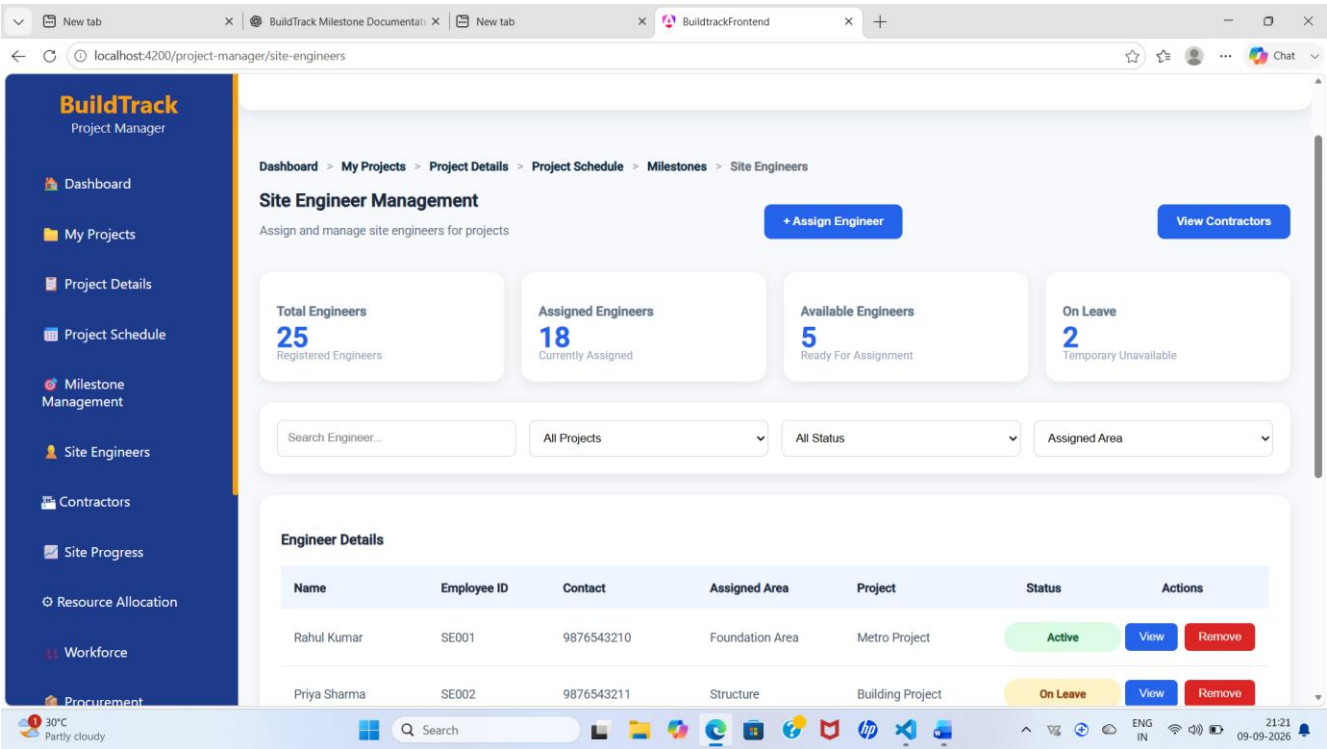


Figure 14: BuildTrack Project Manager – site engineers

Site engineers

6.9 Contractors

The Contractors page allows the Project Manager to view contractors associated with the project. It provides information about contractor companies and their areas of specialization. This helps the Project Manager coordinate and monitor contractor involvement in project activities.

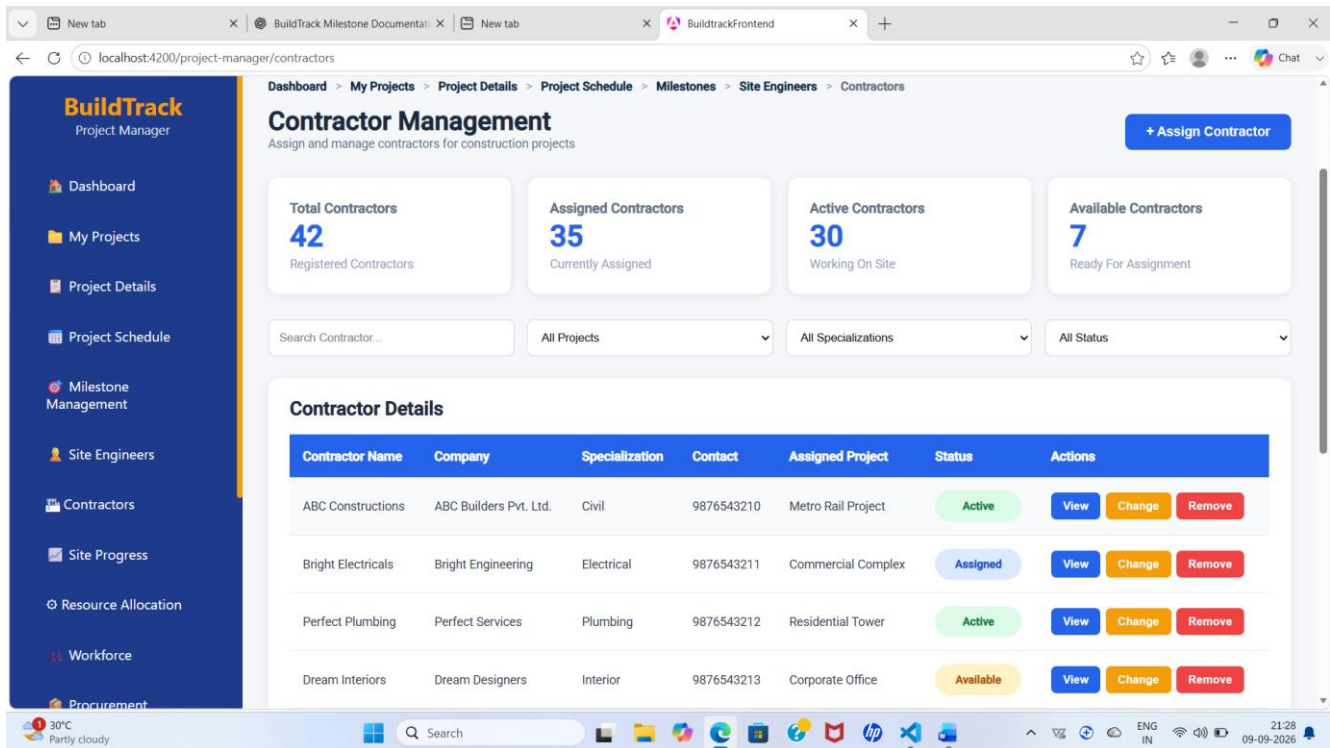


Figure 15: BuildTrack Project Manager – contractors

Contractors

6.10 Site Progress

The Site Progress page allows the Project Manager to monitor the progress of construction activities at the site. It provides an overview of current progress and work completion status. This helps the Project Manager track project execution and identify progress-related issues.

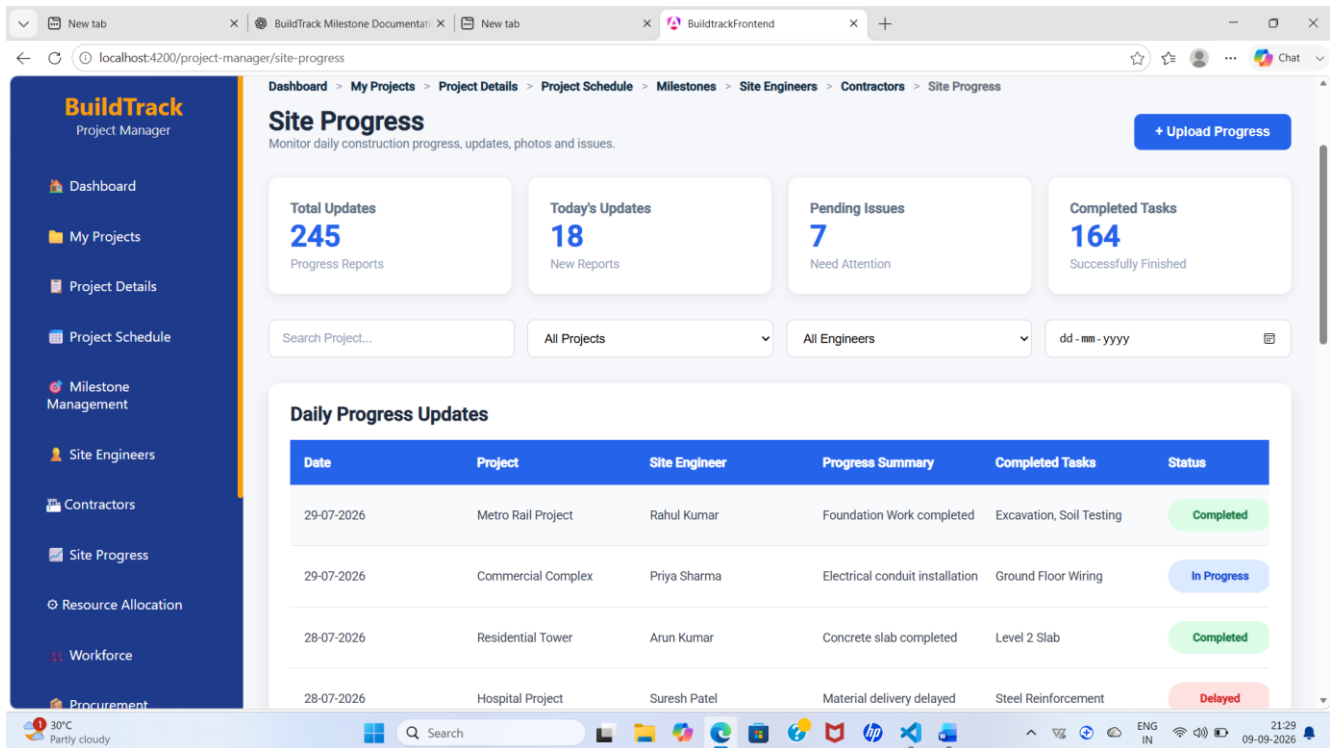


Figure 16: BuildTrack Project Manager – site progress

Site progress

6.11 Resource Allocation

The Resource Allocation page allows the Project Manager to view resources allocated to projects. It provides information about resource assignments and their association with project activities. This helps the Project Manager monitor and coordinate resource utilization.

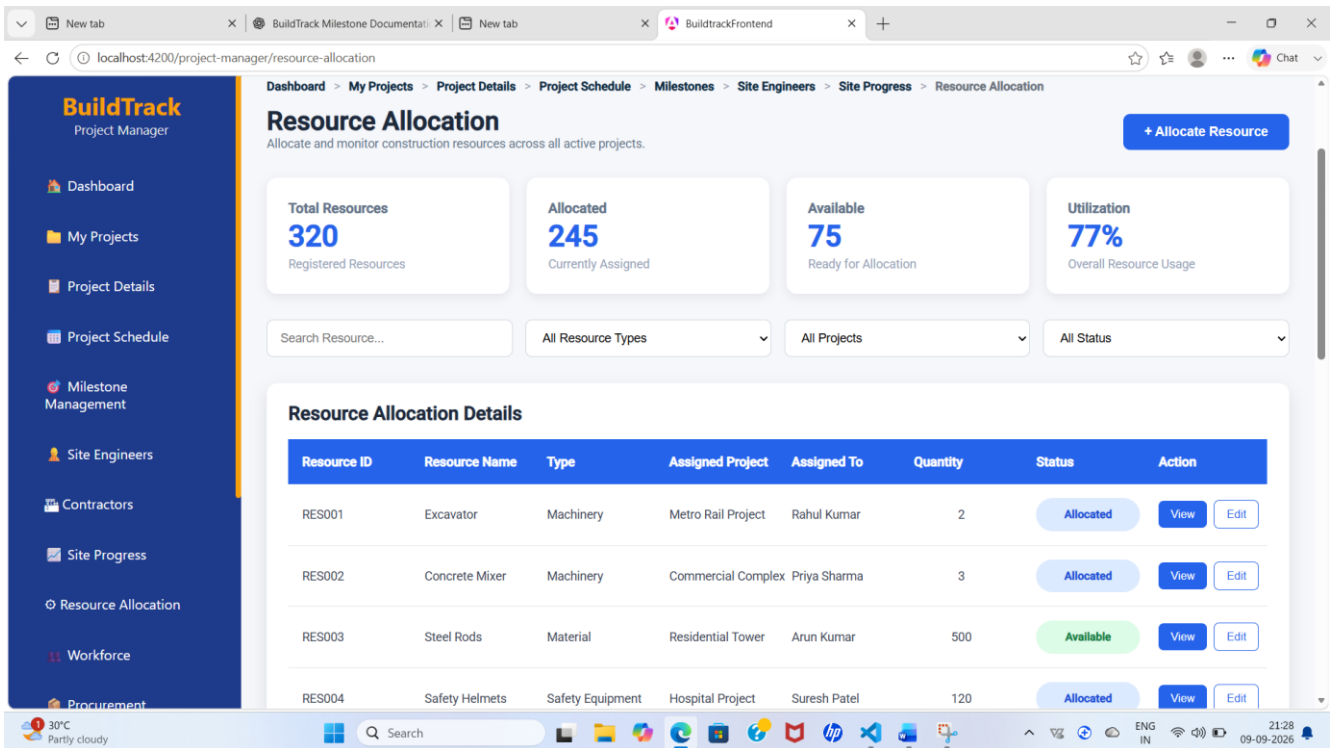


Figure 17: BuildTrack Project Manager – resource allocation

Resource allocation

6.12 Workforce

The Workforce page allows the Project Manager to view workforce and team information associated with projects.

It provides an overview of personnel involved in project activities.

This helps the Project Manager monitor workforce availability and project assignments.

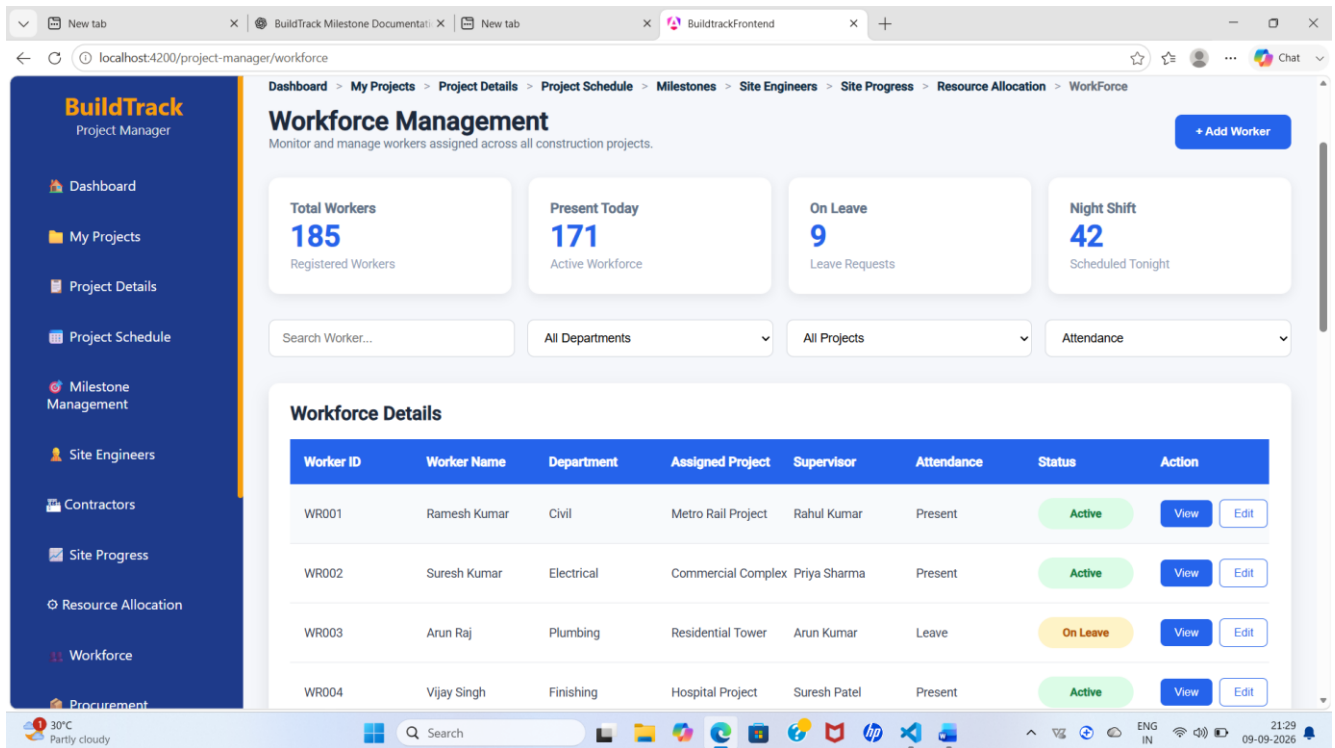


Figure 18: BuildTrack Project Manager – Workforce management

Workforce management

6.13 Procurement Requests

The Procurement Requests page allows the Project Manager to view procurement requests related to project requirements.

It provides information about requested materials, suppliers and procurement status.

This helps the Project Manager monitor material requirements and procurement activities.

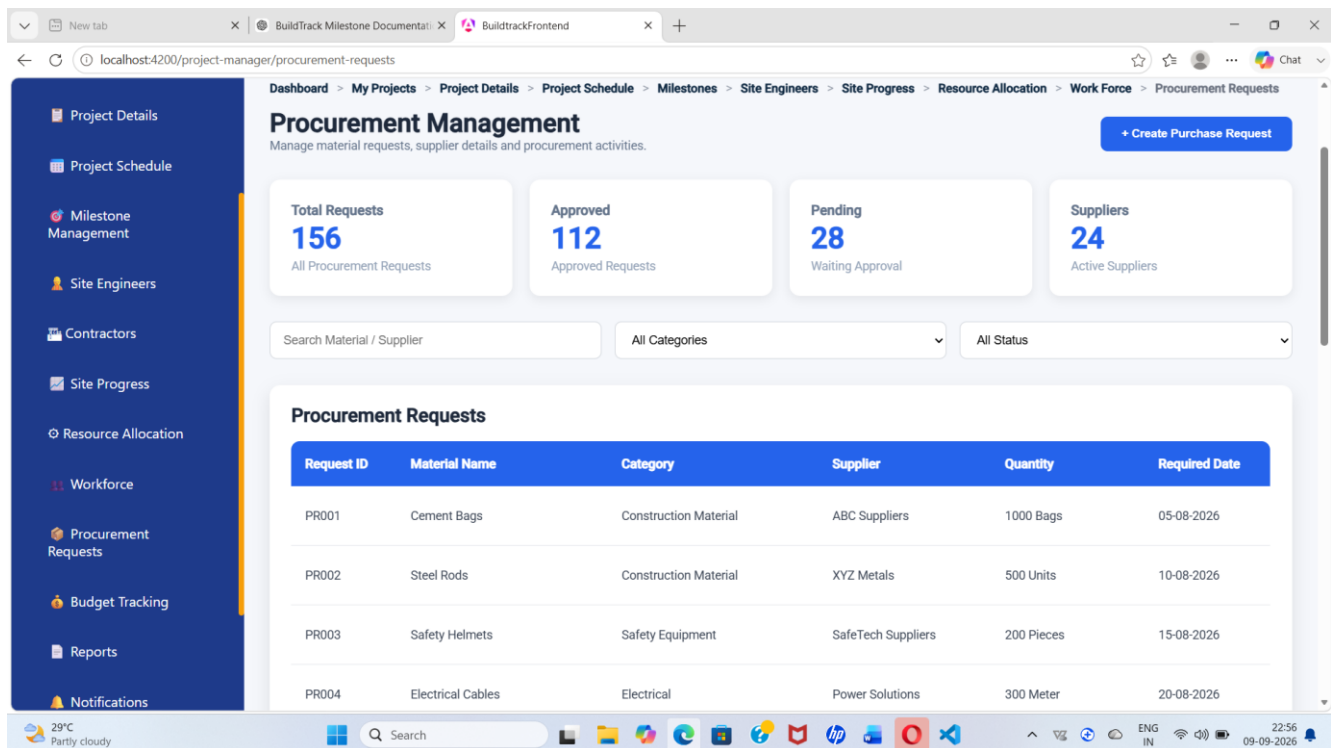


Figure 19: BuildTrack Project Manager – procurement requests

Procurement Requests

6.14 Budget Tracking

The Budget Tracking page allows the Project Manager to monitor the financial status of projects. It provides information about the project budget, spending and remaining amount. This helps the Project Manager track budget utilization and control project costs.

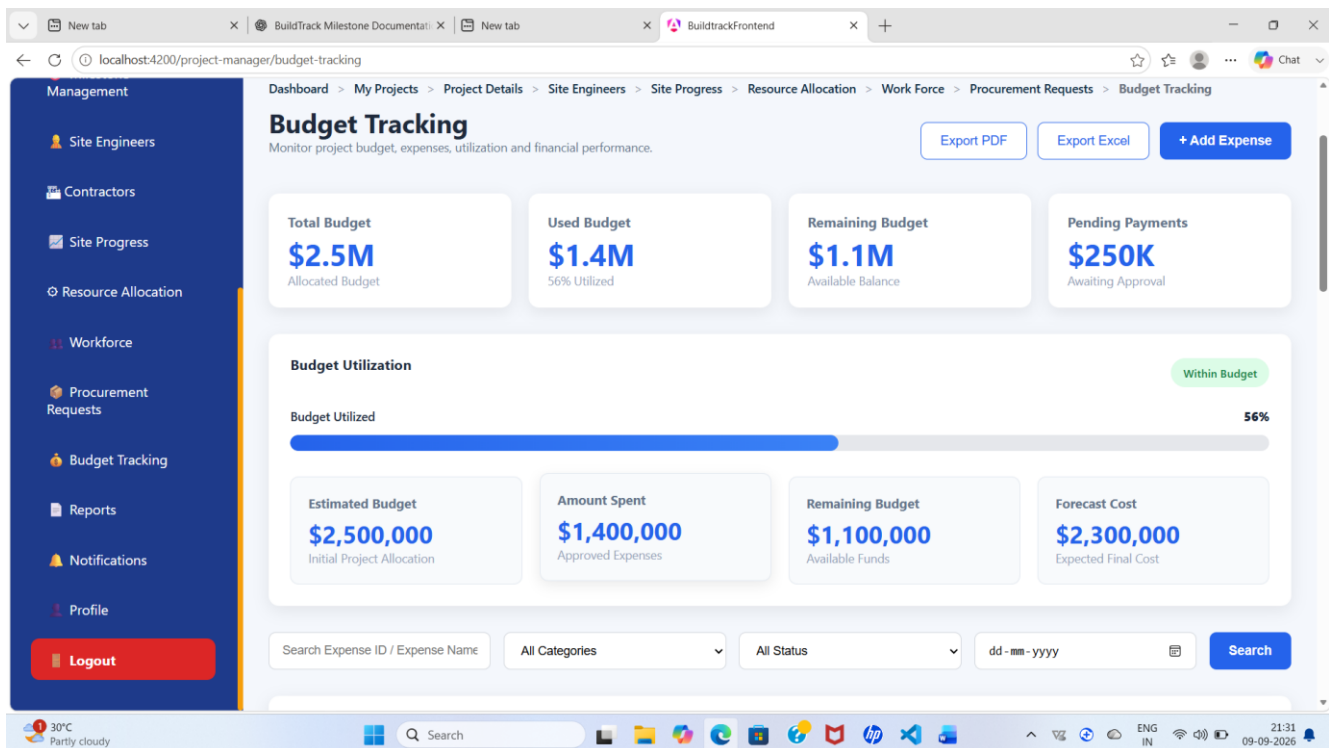


Figure 20: BuildTrack Project Manager – budget tracking

Budget tracking

6.15 Reports

The Reports page provides the Project Manager with access to project-related reporting information. It helps organize and review important project data and performance details. This supports project monitoring and informed decision-making.

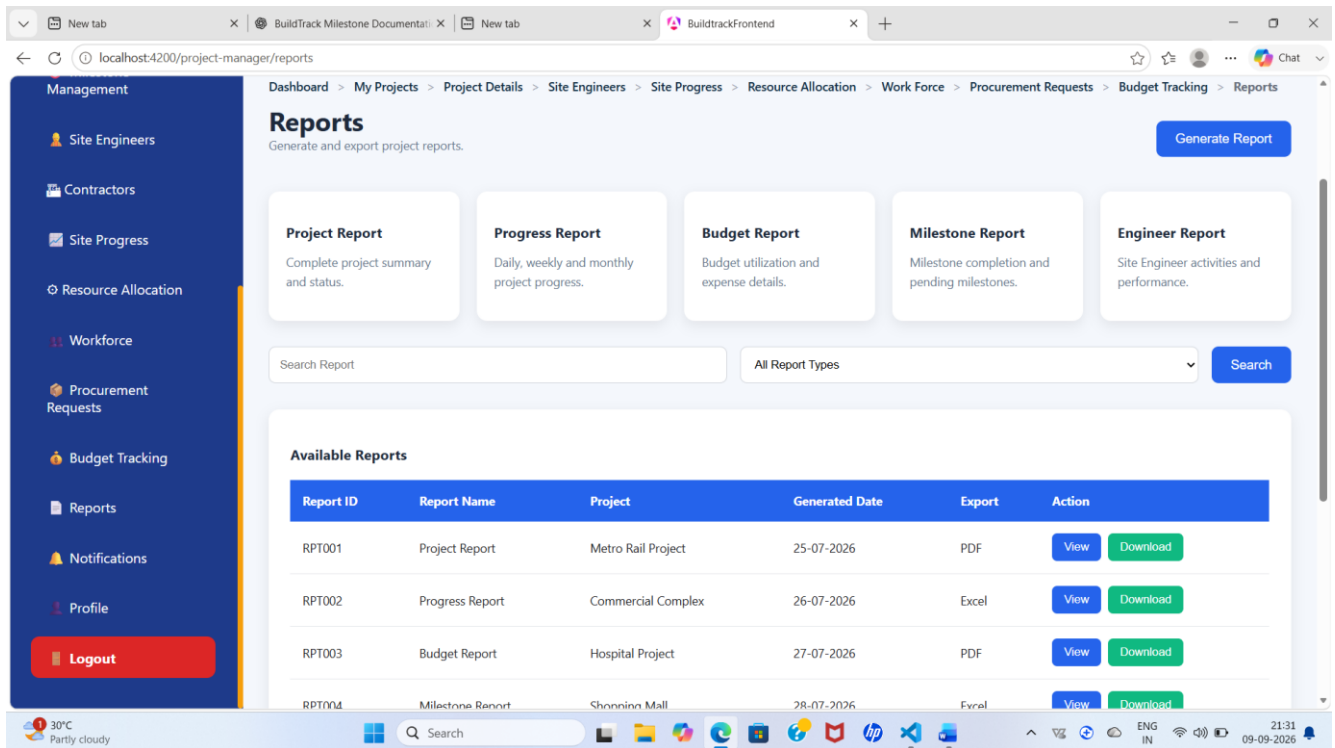


Figure 21: BuildTrack Project Manager – reports

Reports

6.16 Notifications

The Notifications page allows the Project Manager to view project-related notifications and updates. It provides important alerts and information related to ongoing project activities. This helps the Project Manager stay informed about changes and important events.

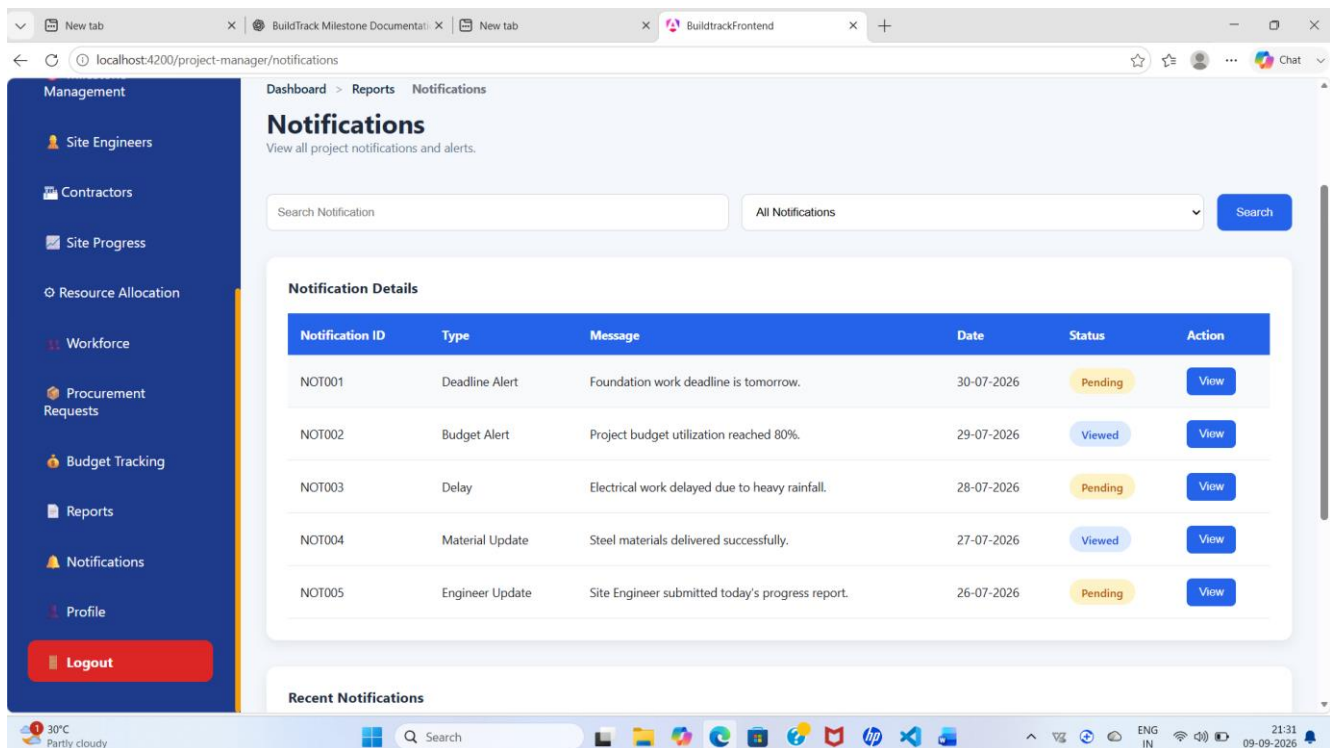


Figure 22: BuildTrack Project Manager – notifications

Notifications

6.17 Profile

The Profile page allows the Project Manager to view their profile information. It displays the user's relevant account and personal information. This provides easy access to Project Manager account details.

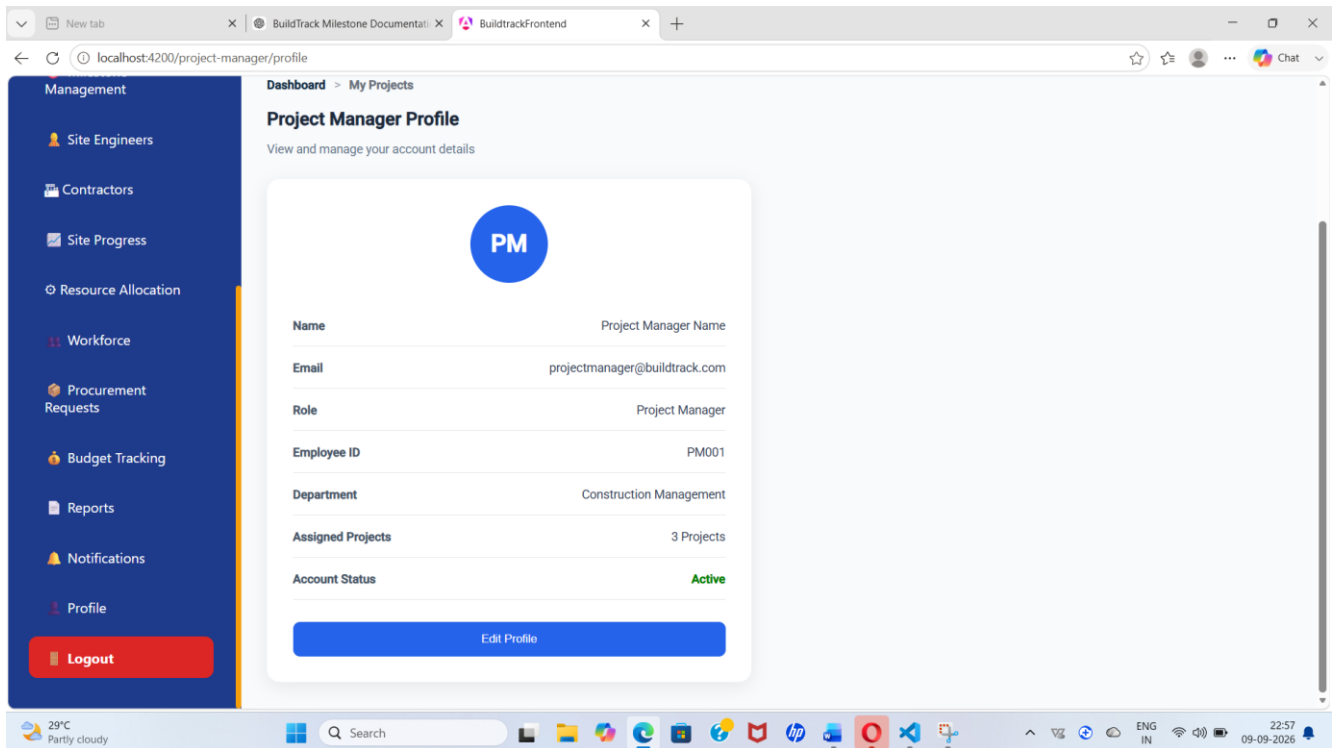


Figure 23: BuildTrack Project Manager – profile

Profile

6.18 Logout

The Logout page allows the Project Manager to securely exit the BuildTrack system. When logout is selected, the system ends the current user session and navigates to the Login page. This helps protect user access and ensures the Project Manager can log in again when required.

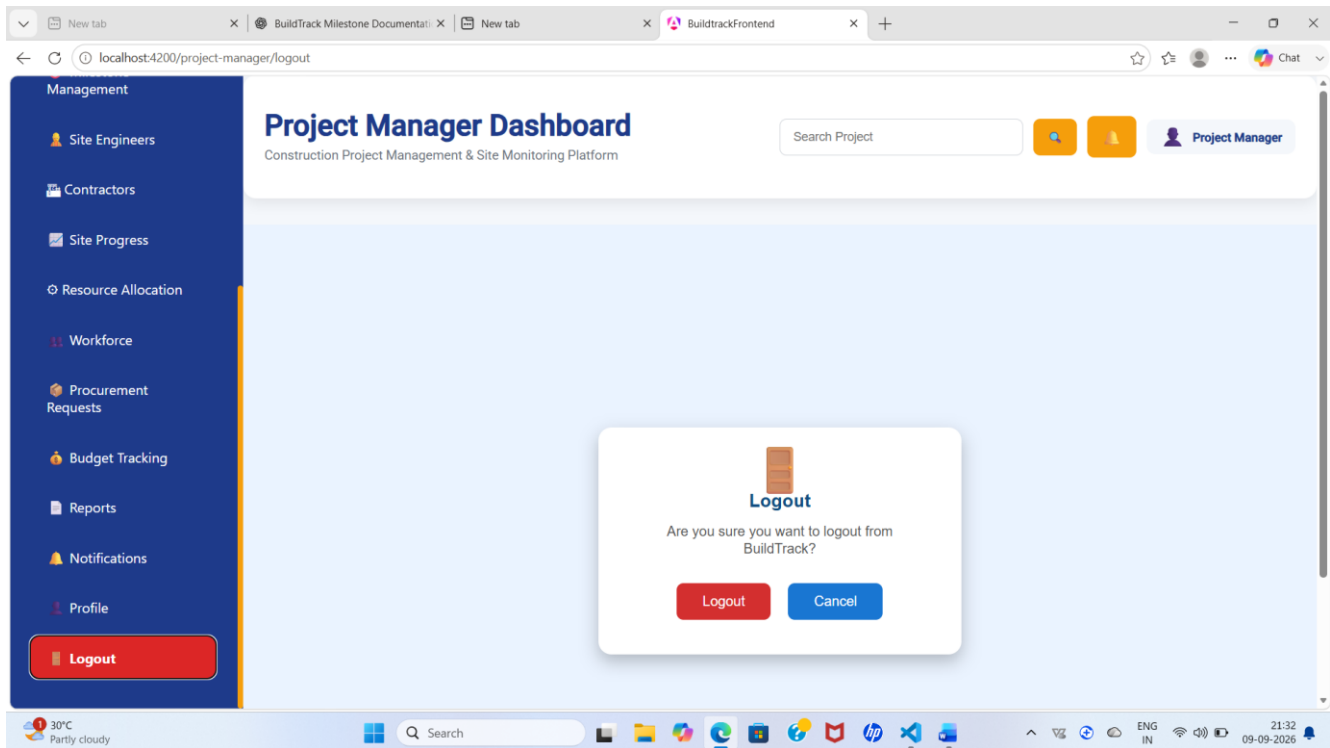


Figure 24: BuildTrack Project Manager – Logout

Logout

6.19 Project Manager Workflow

Step	User Action	System Response	Result
1	Log in as Project Manager.	Authentication response returns the Project Manager role.	User is redirected to the Project Manager Dashboard.
2	Review dashboard summary.	Dashboard displays project, milestone, delay and budget indicators.	Manager obtains a high-level project status.
3	Open My Projects.	Project list and search controls are displayed.	Manager selects a project for review.
4	Open Project Details.	Project information, progress, milestones, engineers, contractors, resources and materials are displayed.	Manager reviews project execution details.
5	Open Schedule/Milestones.	Timeline and activity/milestone tables are displayed.	Manager monitors planning and completion status.
6	Review supporting areas.	Site progress, resource, workforce, procurement and budget views are available.	Manager gets a broader operational picture.
7	Use reports/notifications/profile.	Corresponding sections are opened through Project Manager navigation.	Manager can review supporting project information.

6.20 Module Outcome

The Project Manager workspace provides a broad frontend representation of construction project monitoring and planning. The dashboard and individual project pages are structured around project progress, milestones, resources, workforce, procurement, budget and reporting concepts.

7. Role-wise Functionality

Role	Milestone 1 Capabilities Evidenced
Administrator	Access administrator dashboard; manage/view user information; add/edit/view user screens; manage/view project information; add/edit/view project screens; administrative analytics/report/audit-log routes.
Project Manager	Access Project Manager dashboard; review assigned projects; inspect project details; monitor schedules and milestones; review site progress, engineers, contractors, resources, workforce, procurement and budget information; access reports, notifications and profile.
Site Engineer	The authentication component contains role-based navigation to the Site Engineer dashboard, and protected Site Engineer routes are defined. Detailed Site Engineer functionality belongs to a later milestone and is not documented as part of Milestone 1.
Other Roles	No additional Milestone 1 functionality is claimed unless evidenced by the current source.

8. End-to-End Milestone 1 Workflow

User → Login → FastAPI Authentication → JWT + Role → Angular Role-Based Navigation → Administrator or Project Manager Workspace → Project/User Information → Monitoring and Planning Views → Operational Output

Input	Processing / Action	System	Output
Email + password	Credential validation and password verification	FastAPI authentication service	JWT access token and role
Role	Role evaluation and route selection	Angular routing	Administrator / Project Manager workspace
User data	Create, inspect or edit user information	Admin user-management screens	Updated/visible user information
Project data	Create, inspect or edit project information	Admin project-management screens	Updated/visible project information
Project status data	Review progress, milestones, schedule and supporting operational sections	Project Manager workspace	Project monitoring and planning view

9. Screenshots and Evidence Checklist

- Figure 1 – Login Page
- Figure 2 – Administrator Dashboard
- Figure 3 – Administrator User Directory
- Figure 4 – Add New User
- Figure 5 – View User
- Figure 6 – Administrator Project Management
- Figure 7 – Project Details
- Figure 8 – Add New Project

- Figure 9 – Project Manager Dashboard
- Figure 10 – My Projects
- Figure 11 – Project Details
- Figure 12 – Project Schedule
- Figure 13 – Milestone Management

Use screenshots from the running BuildTrack application rather than code-editor screenshots. Crop out unrelated desktop areas. Keep the same image width throughout the document and place each screenshot immediately after the relevant feature description.

10. Technical Implementation Summary

Area	Implementation Evidence
Frontend Framework	Angular 22 source structure with standalone components.
UI Components	Angular Material modules are used in the Login component; other pages use HTML/CSS and Angular templates.
Routing	Angular route configuration defines authentication, administrator, Project Manager and other module routes.
Backend Framework	FastAPI application with multiple routers.
Authentication	JWT token creation/verification, role included in login token data, password hashing and verification.
Database Layer	SQLAlchemy engine/session configuration and ORM models for users and project-related entities.
CORS	Backend CORS middleware allows local Angular development origins at localhost:4200 and 127.0.0.1:4200.

11. Limitations / Pending Work

- The Register frontend currently contains a placeholder message rather than a complete registration form.
- The Reset Password frontend currently contains a placeholder message rather than a complete recovery UI.
- The generic Admin User Management page itself is a placeholder; the separate User Directory, Add User, View User and Edit User screens are present.
- Some Project Manager pages display predefined frontend data structures; the documentation should not describe these values as live database values unless backend integration is demonstrated.
- The project source contains later-milestone modules, but they are intentionally not described as completed Milestone 1 deliverables.
- Any backend persistence or API integration that is not demonstrable in the running application should be marked as pending rather than claimed as fully operational.

12. Milestone 1 Completion Summary

Item	Status / Evidence
Authentication login	Implemented in Angular and connected to AuthService/backend login flow.
JWT authentication	Backend token creation and verification are implemented.
Role-based navigation	Login response is used to redirect Administrator, Project Manager and Site Engineer users.

Administrator dashboard	Implemented frontend dashboard with KPI and project-monitoring components.
User management screens	User directory, add, view and edit screens are present.
Project management screens	Project listing, details, add, edit and view screens are present.
Project Manager dashboard	Implemented with project, milestone, site, resource, workforce, procurement, budget, delay, analytics, report, notification and profile sections.
Project planning pages	My Projects, Project Details, Project Schedule and Milestones are present.
Pending frontend work	Registration and Reset Password UIs require completion; some pages use frontend/static data.

13. Conclusion

Milestone 1 establishes the foundation of BuildTrack through authentication, administration and Project Manager workspaces. The source code demonstrates the Angular routing structure, authentication service interaction, JWT-based backend security foundation and a set of administrative and project-management interfaces. The remaining limitations are explicitly identified so that the documentation reflects the actual implementation rather than the planned specification alone.